



Indira Gandhi Delhi Technical University for Women
(Established by Govt. of Delhi vide Act 09 of 2012)
Kashmere Gate, Delhi-110006

B.Tech. – Computer Science and Engineering					
Fifth Semester					
S No	Type of Course	Subject	L-T-P	Code	Credits
1.	DCC	Object Oriented Programming	3-0-2	BIT 202	4
2.	DCC	Artificial Intelligence	3-0-2	BAI 202	4
3.	DCC	Theory of Computation	3-1-0	BCS 301	4
4.	DEC	Data Analytics Models and Algorithms	3-0-2	BCS 302	4
		Human Computer Interaction	3-0-2	BCS 303	
		Cryptography	3-0-2	BIT 319	
5.	AEC/VAC	Disaster Management	2-0-0	AEC 301	2
6.	SEC	Cloud Computing Systems and Applications	1-0-4	BCS 304	3
7.	Internship	Internship	-	BCS 350	2
		Total			23
Sixth Semester					
S No	Type of Course	Subject	L-T-P	Code	Credits
1.	DCC	Cyber Security	3-0-2	BIT 320	4
2.	DCC	Machine Learning	3-0-2	BAI 301	4
3.	DCC	Advanced Computer Network	3-0-2	BCS 310	4
4.	DEC 1	Big Data Analytics	3-0-2	BCS 311	4
		Database Warehousing and Business Intelligence	3-0-2	BAI 314	
		Multimedia Technology	3-0-2	BCS 312	
5.	DEC 2	Compiler Design	3-0-2	BIT 321	4
		Advanced Computer Architecture	3-0-2	BCS 313	
		Wireless Sensor Network	3-0-2	BCS 314	
6.	AEC	Innovation, Business Models and Entrepreneurship (By Prof. Rajat Agrawal, Prof. Vinay Sharma, IIT Roorkee)	2-0-0	AEC 302	2
		Project Development on Societal Issues	0-0-4	AEC 303	
		Innovation and Business Entrepreneurship	2-0-2	AEC 304	
		Total			22

OBJECT ORIENTED PROGRAMMING	
Course Code: BIT202	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: II/III
Course Category: SEC/DCC	

Introduction: This course is designed to enable students to understand object oriented concepts, principles and techniques using C++. As well as students will understand how to use C++ classes, class libraries and related concepts like inheritance, polymorphism and File Input output.

Course Objectives:

- To learn Object Oriented principles
- Students will able to practice object oriented programming concepts using C++ programming

Course Outcomes: After completion of the course, the students will be able to:

- CO1:** To learn the process of representing problems and writing, compiling and debugging programs.
- CO2:** To introduce the principles of inheritance and polymorphism.
- CO3:** Implement programs to express proficiency and improve effective programming skills using classes, objects.
- CO4:** Understand commonly used operations for file system, exception handling and create namespace solutions.

Pedagogy: Lectures will be imparted along with hands-on lab sessions and development of Object Oriented programs with use of arrays, pointers, classes and objects.

Contents

UNIT-I	10 Hours
Introduction to Object Oriented Programming: Need for Object Oriented Programming, Comparison of Programming paradigms, Characteristics of Object-Oriented Programming Languages, Structure of a C++ program, Use of cin and cout, Compilation process. C++ Programming Language (Procedural): Tokens, Data Types (Basic, Advanced and Derived), Variables, Reference vs Pointers, Operators (scope resolution, dynamic memory related, typecast), Expressions, Functions (inline function, const arguments, default arguments).	
UNIT-II	11 Hours
Classes and Objects: Objects, Classes, Encapsulation, Data Abstraction, Role of private and public access specifier, Memory organization of class, Member functions – inline and non inline, static member variables, Friend functions, Class vs Structure, Constructors – default, parameterized, copy and dynamic, Destructors, Assignment operator Polymorphism: Function overloading, Constructor overloading, Compile time polymorphism, Overloading Rules, Operator Overloading (Unary and Binary) as member function/friend function, Example operators to be overloaded: Arithmetic, Output/Input, Prefix/ Postfix Increment and Decrement, Comparison, Assignment, subscript and function call Operator.	
UNIT-III	10 Hours
Inheritance: Inheritance, Types of Inheritance, Use of protected access specifier, Virtual base class, Ambiguity resolution using scope resolution operator and Virtual base class, Overriding inheritance methods, Constructors and Destructor in derived classes. Runtime polymorphism, Pointer to objects, Virtual Functions (concept of virtual table), pure virtual functions, Abstract Class. Managing Input / Output: Concept of streams, console I/O, File I/O – Predefined classes, file opening & closing, file manipulation, read & write operations, sequential and random file access.	

UNIT-IV		11 Hours
Exception Handling: Basic mechanism, Throwing, Catching and Re-throwing. Namespace: Basic concept, role of scope resolution operator and using keyword. Generic Programming: User defined Templates - Class templates with and without multiple parameters and Function templates with and without parameters, Template overloading.		
Text Books		
1.	Bjarne Stroustrup , "The C++ Programming Language", Addison Wesley, 4th Edition, 2022/latest edition	
2.	Herbert Schildt, "C++: The Complete Reference", 4th Edition, McGraw Hill, 2017/latest edition.	
Reference Books		
1.	Balagurusamy, E. "Object oriented programming with C++.", 8th Edition, McGraw Hill, 2021 /latest edition	

ARTIFICIAL INTELLIGENCE	
Course Code: BAE 202	Credits: 4
Contact Hours: L-3 T-0 P-2	Semesters: 4
Course Category: DCC	

Introduction:

This course is an introduction to the basic knowledge representation, problem solving and learning methods in the field of artificial intelligence. After completing this course, students should be able to understand the basic concepts of problem solving and learning.

Course Objective:

- To introduce the basics of artificial intelligence, including intelligent agents, problem-solving methods, and search strategies.
- To explore how knowledge is represented and processed using logic, rules, and text features.
- To understand how to handle uncertainty using probability, fuzzy logic, and neural networks for decision-making.
- To learn about real-world AI applications like expert system.

Prerequisite:

Discrete Mathematics, Programming Concepts.

Course Outcomes:

After successful completion of the course; the student will be able to:

CO1: Explain the principles of intelligent agents, search strategies, and the Turing Test, and apply BFS and DFS algorithms to solve search-based problems.

CO2: Implement knowledge representation techniques.

CO3: Differentiate and evaluate uncertainty handling techniques.

CO4: Apply the concepts of expert systems and evaluate applications of AI in diverse fields.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

Contents

UNIT-I	10 Hours
Introduction to AI: Brief introduction about Intelligent agents and Problem Solving. Turing Test. Uninformed Search Strategies, Informed Search Strategies, Heuristics. Solving problems by searching, BFS, DFS, Issues in design of Intelligent Search Algorithms.	
UNIT-II	10 Hours
Knowledge Representation: Knowledge Representation using predicate logic, Rule Based Systems, Ontology, WordNet and Concept Net as Knowledge representation tools Programming with Prolog/Lisp. Text Feature Extraction - BoW Model, TF-IDF. Word Embeddings - Word2Vec, GloVe.	
UNIT-III	12 Hours
Decision Making in Uncertainty: Handling Uncertainty, Probabilistic Reasoning, Fuzzy Logic, Learning by induction, Introduction to Neural Network Genetic Algorithms basics. Rough Sets. Case Studies of Applications of Uncertainty	
UNIT-IV	10 Hours
Real World Applications of AI: Real World Applications of AI: Expert System Architecture, Case Studies: MYCIN, Applications in NLP, Medical Sciences, Social Network Analysis, Information Retrieval from Search Engines and Metasearch Engines, IoT Applications & Big Data Analytics Applications. Ethics in AI.	
Text Books	
1.	S.J. Russell and P. Norvig, "Artificial Intelligence- A Modern Approach", Pearson 4 th Edition, 2022/Latest Edition.
2.	P.H. Winston, "Artificial Intelligence", Pearson Education, 3 rd Edition, 2002/ Latest Edition.
Reference Books	
1.	E. Kumar, "Artificial Intelligence", Dreamtech Press, 2020/ Latest Edition.
2.	E. Rich and K. Knight, "Artificial Intelligence", McGraw Hill Education; 3 rd Edition 2017, Latest Edition.
3.	L. Luger, "Artificial Intelligence : Structures and Strategies for Complex Problem Solving", Pearson Education, 5 th Edition 2008/ Latest Edition.
4.	N.J. Nilsson, "Principles of Artificial Intelligence", Narosa Publ. House, 2002/ Latest Edition.

Subject: Theory of Computation

Course Code: BCS 301
Contact Hours: L-3 T-1 P-0
Course Category: DCC

Credits: 4
Semester: 5

Introduction: The study of automata and the theory of computation deal with the concepts of working of automatic machine and processing of input formal language data. This subject provides an important background material to students involved in understanding the basic functionalities of automata theory.

Course Objective:

- To introduce concepts in Automata theory and theory of computation
- To introduce different formal language classes and their relationships
- To introduce grammars and recognizers for different formal languages

Pre-requisite: Basic concepts of mathematics

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 To Understand properties of formal languages, automata, their equivalence, conversion techniques, concept of Context Free Grammars, and Pushdown Automata.

CO2 Understanding of the key results in algorithmic complexity, computability and solvability of problems.

CO3 To Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.

CO4 Analyze the finite automata and regular expressions for accepting the language

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I

10 hours

Introduction to Theory of Computation: Definitions: Languages, Grammar, Automata, Applications of Theory of Computation, **Finite Automata:** DFA, NDFA, Equivalence of DFA and NDFA, DFA Minimization, Regular Languages, Regular Expression, Properties of Regular Languages, Pumping Lemma of Regular Languages.

UNIT II

10 hours

Context Free Language: Introduction Context Free Grammar, Ambiguity, Chomsky Normal Form(CNF), Greibach Normal Form(GNF), Derivation Tree, Derivation types, Pushdown Automata (PDA), Non Deterministic PDA, Equivalence of PDA and CFGs, Pumping Lemma for context free languages.

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UNIT III

10 hours

Pumping Lemma: Turing Machine: Definition, TM as language acceptors, TM as transducers, Hierarchy of Formal Languages and Automata, Chomsky Hierarchy, Context Sensitive Languages and LBA, Unrestricted Grammars.

UNIT IV

10 hours

Finite State Transducers: Introduction, Mealy Machines, Moore Machines, Mealy and Moore Equivalence, Limitations of Finite State transducers. **Undecidability:** Undecidable Problem, Posts Correspondence Problem, the Classes P and NP, NP Complete Problem, Halting problem.

Text Books

1. P. Linz "An Introduction to Formal Languages and Automata", Narosa Publishers, 2010.
2. J. Ullman, J. Hopcroft "Introduction to Automata Theory, Languages and Computation", Pearson Education India; 3rd edition (2008).

Reference Books

1. M. Sipser "Introduction to the Theory of Computation", Cengage; 3rd edition (2014)
2. C.K. Nagpal "Formal Languages and Automata Theory", Oxford University Press (2015)

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DATA ANALITICS MODELS AND ALGORITHMS

Course Code: BCS-302
Contact Hours: L-3 T-0 P-2
Course Category: DEC

Credits:4
Semester: 6th

Introduction:

Data analysis plays an important role in extracting meaningful information from raw data. With the growth of digital technologies, large volumes of data are generated in various fields such as healthcare, finance, education and business. Proper collection, cleaning and analysis of data helps organizations make better decisions. Algorithms are widely used to process, classify and group data efficiently. This course introduces students to fundamental concepts of data analysis, statistical methods and basic algorithms such as regression, classification and clustering. It provides knowledge about data preprocessing, visualization and model evaluation techniques used for analyzing datasets.

Course Objectives:

At the end of the course students should demonstrate the ability to collect, process and analyze data using suitable statistical methods and algorithms.

Pre-Requisites:

Essential: Basic Statistics, Fundamentals of Programming

Desirable: Database Management Systems (DBMS)

Course Outcomes:

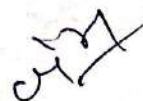
CO1: Perform data collection and preprocessing from different data sources.

CO2: Apply descriptive statistical measures such as mean, median, mode and harmonic mean for data analysis.

CO3: Use visualization techniques and correlation methods to analyze patterns in datasets.

CO4: Implement basic algorithms such as regression, classification and clustering for prediction and grouping of data.

Pedagogy:



The course will be delivered through lectures supported by practical demonstrations and problem-solving sessions. Tutorial work shall include pen and paper exercises related to statistical measures and algorithm steps. Practical assignments will involve data cleaning, visualization and implementation of simple algorithms such as classification and clustering using appropriate tools. Students will be encouraged to practice independently using datasets provided during lectures. There shall be about 4–5 hours per week of take-home work based on lecture content.

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UNIT 1

Introduction to Data Analysis: data analysis, importance, evolution. Steps involved in data analysis

Types of Data: Structured data, unstructured data, qualitative data, quantitative data.

Applications of Data Analysis: Use of data analysis in healthcare, finance, education, retail, sports.

Data Ethics and Security: Data privacy, data protection, security issues in data handling.

UNIT 2

Data Collection: Primary data collection, secondary data collection, surveys, observations, online sources. Data Pre-Processing: Overview of preprocessing, importance of data preparation before analysis.

Data Cleaning: Handling missing values, removing duplicate data, correcting inconsistent data. Data Integration and Transformation: Combining data from multiple sources, data transformation methods, normalization techniques.

Data Reduction and Discretization: Reducing data size, feature selection basics, converting continuous data into discrete values

UNIT 3

Descriptive Statistics: Measures of central tendency, mean, median, mode, harmonic mean. Measures of Dispersion: Range, variance, standard deviation. Data Distribution: Skewness, kurtosis, understanding data distribution.

Data Visualization Techniques: Histogram, box plot, scatter plot, heat map, pivot table.

UNIT 4

Algorithms in Data Analysis: Basic concept of algorithms, role of algorithms in data analysis. Regression Techniques: Simple linear regression, multiple regression, logistic regression.

Classification Algorithms: k-Nearest Neighbors (KNN), decision tree classification, prediction techniques. Clustering Algorithms: k-means clustering, hierarchical clustering.

Model Evaluation: Confusion matrix, accuracy, precision, recall, F1 score, cross validation, overfitting and underfitting.

Text Books

1. Thomas Runkler, Data Analysis Models and Algorithms for Intelligent Analatics.

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2. Han J., Kamber M. and Pei J., Data Mining: Concepts and Techniques, Morgan Kaufmann/Latest Edition

3. Zaki M.J. and Meira W. Jr., Data Mining and Analysis: Fundamental Concepts and Algorithms, Cambridge University Press/Latest Edition

4. Maheshwari A., Data Analytics Made Accessible, Wiley/Latest Edition

Reference Books

1. McKinney W., Python for Data Analysis, O'Reilly Media/Latest Edition

2. Cormen T.H., Leiserson C.E., Rivest R.L. and Stein C., Introduction to Algorithms, MIT Press/Latest Edition

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Subject: Human Computer Interaction

Course Code: BCS 303
Contact Hours: L-3 T-0 P-2
Course Category: DEC

Credits: 4
Semester: 5

Introduction: HCI is a multidisciplinary course focusing on design of computer technology interaction between humans particularly users, and computers. HCI has expanded and evolved and now covers several forms of information technology design.

Course Objective:

- To Impart knowledge and understanding about human cognition and human perspective with respect to computers.
- To Introduce User Interface and Graphical User Interface elements
- To introduce components and software tools for Human computer interaction.

Pre-requisite: Computer fundamentals

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 To understand the importance of good user interface design and the difference between UI and GUIs

CO2 To understand and apply the design process of UIs

CO3 To understand and apply screen-based UI principles

CO4 To apply the UI design and interaction-based device components to UI designing

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped classroom teaching will be adopted.

UNIT I

10 hours

Introduction: Importance of user interface, definition, importance of good design, Benefits of good design, A brief history on screen design. **Graphical User Interface:** Popularity of Graphics, the concept of direct manipulation, graphical system, characteristics, web user interface popularity, characteristics-principles of user interface.

UNIT II

11 hours

Design process: Human interaction with computers, importance of human characteristics, human considerations, human interaction speeds, understanding business junctions. Limitations and constraints of mobile and hand-held platforms and issues related to network, power, memory and UI responsiveness required with reference to other computing platforms.



UNIT III

11 hours

Screen Designing (UI Design): Design goals, screen planning and purpose, organizing screen elements, ordering of screen data and content, screen navigation and flow, visually pleasing composition, amount of information, focus and emphasis, presenting information simply and meaningfully, information retrieval on web, statistical graphics, technological consideration in interface design. **Windows:** Navigation schemes selection of window, selection of devices based and screen-based controls.

UNIT IV

10 hours

Components and Software Tools: Components: text and messages, Icons and increases, multimedia, colors, user problems. **Software tools:** Specification methods interface, building tools. **Interacting Devices:** Keyboard and function keys, touch screen technology, pointing devices, speech recognition digitization and generation, image and video displays (LED, LCD, HD displays), voice activated commands and its implementation

Text Books

1. W. O Galitz, "The essential guide to user interface design", Wiley, Third Edition, 2007.
2. B. Shneidermann, "Designing the User Interface: Strategies for Effective Human Computer Interaction", 5th Edition, Pearson Education, 2014.

Reference Books

1. A. Dix, J. Finck, G. Goryd, A. Boud, R. Beaulieu, "Human Computer Interaction", 3rd Edition, Pearson Education, 2004.
2. Sharp, Rogers, Preece, "Interaction Design Beyond Human-Computer Interaction", 5th Edition, Wiley, 2019
3. D. Stone, C. Jarrett, M. Woodroffe, and S. Minocha, "User Interface Design and Evaluation(interactive technologies) , Morgan Kaufmann; Revised of "The Miracle of Col ed. edition (2005)

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Cryptography

Course Code: BIT- 319
Contact Hours: L-3 T-0 P-2
Course Category: DEC/DCC

Credits: 4
Semester: 5

Introduction:

This course will introduce students to basic building blocks of cryptography and applications of cryptographic protocols in real world. The focus will be on how cryptography and its application can maintain privacy and security in electronic communications and computer networks.

Course Objective:

- To understand the fundamentals of Cryptography
- To acquire knowledge on standard algorithms used to provide confidentiality, integrity and authenticity
- To explain and use modern cryptographic methods (symmetric encryption, public key encryption, hash functions, key management, digital signatures, certificates)
- To discuss electronic mail security, SSL/TLS and recent developments affecting security and privacy on the Internet.

Pre-requisite: None

Course Outcome:

CO1: Understand applied cryptographic basics.

CO2: Analyze and differentiate between public-key and private key cryptosystems.

CO3: Evaluate security mechanisms using rigorous approaches by key ciphers and hash functions.

CO4: Design cryptographic protocols to solve real world problems.

Pedagogy:

Emphasis on lab sessions where students will be given programming assignments to code in lab based on topics learnt in previous lectures.



Contents

UNIT-I	10 Hours
Course Introduction and terminology, Conventional Cryptography: Definitions, Classical encryption techniques, One time pad, Perfect Secrecy, DES, Triple DES, Finite fields, AES, Modes of Encryption	
UNIT-II	11 Hours
Asymmetric Cryptography: Number Theory, public key cryptography: RSA, ElGamal, and Elliptic Curve Cryptography, Diffie Hellman Key management, Digital Certificates: X.509	
UNIT-III	11 Hours
Stream Ciphers, LFSR based stream ciphers, Message Authentication Codes, Hash functions, Hash algorithms, Digital Signatures and Authentication Protocols, Firewalls	
UNIT-IV	10 Hours
Intrusion Detection, PGP, S/MIME, Key Management, Kerberos, IPSec, SSL/TLS, Password Hashing and Management	

Text Books

1	W. Stallings, "Cryptography and Network Security: Principles and Practice, 8/e", Prentice Hall, 2022/Latest Edition
2	B. Forouzan, D. Mukhopadhyay, "Cryptography and Network Security 3/e", McGraw Hill/Latest Edition
3	Bernard Menezes, "Network Security and Cryptography 2/e", Cengage Learning, 2012/Latest Edition.

Reference Books

1	A. Menezes, P. Van Oorschot, S. Vanstone. "Handbook of Applied Cryptography, 2/e", CRC press, 2018/Latest Edition.
2	Douglas R. Stinson, "Cryptography: Theory and Practice 3/e", CRC Press, 2006/Latest Edition
3	B. Schneier. "Applied Cryptography", 20th Anniversary Edition, John Wiley & Sons, Inc., 2015/Latest Edition



CYBER SECURITY	
Course Code: BIT 320	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: 6
Course Category: DCC/DEC	

Introduction:

This course provides a foundational understanding of cyber security principles, threat landscapes, and defense mechanisms. It introduces students to various types of cyber attacks, cybercrimes, and security vulnerabilities across networks, systems, and applications. The course covers essential security technologies such as cryptography, authentication mechanisms, and secure communication protocols. It also emphasizes security governance, risk management, and legal frameworks relevant to cyber space. The course integrates theoretical concepts with practical awareness of real-world security challenges. It prepares students for advanced studies and entry-level roles in cybersecurity domains.

Course Objectives:

- To develop understanding of cyber security concepts, threats, vulnerabilities, and attack methodologies
- To provide knowledge of various types of cybercrimes and their impact on individuals and organizations
- To introduce security mechanisms such as cryptography, authentication, and secure communication protocols
- To enable understanding of security controls, defense strategies, and risk management practices
- To familiarize students with cyber laws, standards, and governance frameworks

Pre-requisites:

Knowledge of Computer Networking, Linux, UNIX, Understanding of Web Application Architecture and HTTP/HTTPS communication.

Course Outcomes:

After completion of the course the students will be able to:

CO1: Explain cyber security fundamentals, threat landscape, attack models, and basic threat intelligence concepts

CO2: Analyze cybercrimes, application security vulnerabilities, and attack methodologies in modern digital systems

CO3: Apply security mechanisms, cryptographic techniques, and system protection strategies in networked environments

CO4: Evaluate security controls, monitoring frameworks, risk management practices, and cyber laws for securing organizational infrastructure

Pedagogy:

The teaching-learning process will be delivered through lectures, tutorials, assignments, quizzes, and case-based discussions. Emphasis will be placed on real-world cyber attack scenarios, threat analysis, and security mitigation strategies. Assignments will focus on problem-solving and practical understanding of security mechanisms and tools. The course will integrate theoretical concepts with applied learning. Instruction will utilize ICT tools, web-based resources, and board-based teaching for effective knowledge delivery.

Contents

UNIT-I	12 hrs
Introduction and Threat Landscape: Cyber security concepts, security goals and services, threat landscape and attack surface, types of attackers and vulnerabilities, malware – virus, worm, trojan, ransomware, spyware, social engineering attacks – phishing, spear phishing, baiting, network attacks – DoS, DDoS, session hijacking, ARP spoofing, advanced persistent threats (APT), cyber kill chain, basics of threat intelligence – indicators of compromise (IoCs), introduction to anonymity and privacy technologies – proxy servers, VPN, Tor, surface, deep and dark web	
UNIT-II	10 hrs
Cyber Crime and Application Security: Types of cybercrime and attack methodologies, financial frauds – digital payments and UPI frauds, identity theft and account compromise, password attacks and cracking techniques, cybercrime in social media, cyberstalking and online harassment, cloud and mobile security issues, introduction to application security – OWASP Top 10 vulnerabilities, SQL injection, cross-site scripting (XSS), basics of secure coding practices, overview of digital evidence and cybercrime investigation fundamentals	
UNIT-III	10 hrs
Security Mechanisms and System Protection: Device and system security – endpoint, mobile and browser security, operating system and database security basics, cloud security fundamentals – shared responsibility model, identity and access management (IAM), introduction to container security, cryptography – symmetric and asymmetric encryption, key management, hash functions, digital signatures, authentication mechanisms – multi-factor authentication, biometrics, secure communication protocols – SSL/TLS, IPsec, SSH, email security – PGP, S/MIME, wireless security – WPA2/WPA3, introduction to zero trust architecture and endpoint detection concepts	
UNIT-IV	10 hrs
Defense, Monitoring and Governance: Security controls and countermeasures, access control models – DAC, MAC, RBAC, defense-in-depth strategies, firewalls, IDS, IPS, VPN, network access control (NAC), security operations center (SOC) concepts, SIEM basics – log collection, correlation and alerting, threat detection and response basics, honeypots and deception techniques, risk management process, security policies and compliance, cyber laws – Indian IT Act, data privacy principles (GDPR overview), standards and frameworks – ISO/IEC 27001, NIST, COBIT, basics of incident response and business continuity planning	
Text Books	
1. William Stallings, Lawrie Brown, “Computer Security: Principles and Practice”, Pearson Education, 4th Edition, 2018 / Latest Edition	
2. Mark Rhodes-Ousley, “Information Security: The Complete Reference”, McGraw-Hill Education, 2nd Edition, 2013 / Latest Edition	
3. Matt Bishop, “Computer Security: Art and Science”, Addison-Wesley Professional, 2nd Edition, 2018/ Latest Edition	
Reference Books	
4. William Stallings, “Cryptography and Network Security: Principles and Practice”, Pearson Education, 8th Edition, 2023/ Latest Edition	
5. Nina Godbole, Sunit Belapure, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley India, 2nd Edition, 2021/ Latest Edition	



Machine Learning	
Course Code: BAI- 301 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semester: 5

Introduction:

This course provides a concise introduction to the fundamental concepts in machine learning and popular machine learning algorithms. This course will cover the standard and most popular supervised learning algorithms along with the basic clustering algorithms. The course will be accompanied by hands-on problem solving with programming sessions.

Course Objective:

- To understand the problems and difficulties in machine learning.
- To study the strengths and weaknesses of machine learning techniques.
- To gain insights of the supervised and unsupervised learning.
- To apply machine learning approaches for solving real world problems.

Prerequisites: Calculus, Linear algebra, Probability and statistical concepts, Coding and comfort with data manipulation.

Course Outcomes: Upon successful completion of the course, students will be able to:

CO1: Interpret the underlying problems and difficulties that machine learning faces, such as data, model selection, complexity of the model, etc.

CO2: Discuss the strengths and weaknesses of many popular machine learning approaches.

CO3: Analyze the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and unsupervised learning.

CO4: Design and implement various machine learning algorithms in a range of real-world applications.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

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CONTENTS

UNIT I	10 hours
Introduction: Goals and applications of machine learning, Types of Machine Learning: Supervised Learning, Unsupervised Learning, Machine Learning Cycle: Train-Test Split, Validation Data, K- Fold Cross Validation, Evaluation Metrics. Data Exploration and Pre- processing: Data Objects and Attributes; Statistical Measures, Visualization, Data Cleaning and Integration. Feature Extraction and Reduction.	
UNIT II	11 hours
Supervised Learning Regression: Least Mean Square Regression; Ridge Regression and LASSO regression; Logistic Regression, Support Vector Machines, Kernels for learning non-linear functions, K-nearest-neighbor, Bayesian and Naïve Bayes Classifier, Decision Tree Learning.	
UNIT III	11 hours
Unsupervised Learning Learning from unclassified data. Clustering. Hierarchical Agglomerative Clustering, k-means partitional clustering, Hierarchical, and Density-based Clustering, Expectation maximization (EM) for soft clustering. Dimensionality Reduction: Linear Discriminant Analysis; Principal Component Analysis;	
UNIT IV	10 hours
Advanced Topics Measuring the accuracy of learned hypotheses. Comparing learning algorithms: cross-validation, learning curves, and statistical hypothesis testing, Ensemble Learning: Bagging, boosting, and stacking, Random Forests, Ensemble Classification including Adaboost, Active learning with ensembles.	
Text Books	
1	Han, J., Pei, J. and Tong, H., 2022. Data mining: concepts and techniques. Morgan Kaufmann 4 th Edition/ Latest Edition
2	Géron, A., 2022. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media. 3rd Edition/ Latest Edition
3	Daumé, H. III, "A Course in Machine Learning", 2015/ Latest Edition.
4	Mitchell, T. "Machine Learning", 1997/ Latest Edition
Reference Books	
1	Shai Shalev-Shwartz and Shai Ben-David. "Understanding Machine Learning: From Theory to Algorithms", Cambridge University Press, 2014/ Latest Edition
2	Marsland, S., 2014. Machine learning: an algorithmic perspective. Chapman and Hall/CRC 2 nd Edition/ Latest Edition

BTech 5th semester Syllabus

Advanced Computer Network	
Course Code: BCS 310 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semester: 6

Introduction: This course introduces the fundamental concepts of computer networks, including Internet architecture, protocol models, and addressing mechanisms. It provides understanding of routing, transport, and application layer protocols, along with advanced networking concepts such as VPNs, MPLS, and real-time communication. The course also covers wireless networks, mobile IP, and mobile ad hoc networks (MANETs) with their underlying protocols and applications.

Course Objective:

- To understand Internet architecture, networking models, protocols, and addressing schemes (IPv4/IPv6).
- To analyze routing, transport, and application layer protocols for efficient data communication.
- To explore advanced, wireless, and ad hoc networking concepts including VPNs, MPLS, VoIP, and MANETs.

Pre-requisite: Students should have basic knowledge of wireless communication and computer networks.

Course Outcome: Upon successful completion of this course, students will be able to:

CO1: Explain Internet architecture, networking models, and addressing mechanisms including IPv4/IPv6.

CO2: Analyze routing techniques and protocols including intra-domain and inter-domain routing.

CO3: Evaluate transport and application layer protocols for reliable data communication.

CO4: Understand advanced networking concepts such as VPNs, MPLS, and real-time communication (VoIP, SIP).

CO5: Apply concepts of wireless networks and mobile ad hoc networks (MANETs) and their routing protocols.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.



UNIT I		12 hours
Introduction: Structure of Internet, Intranet, Role of ISP and ICANN. Foundation of Networking Protocols: 5-layer TCP/IP Model, 7-Layer OSI Model, Internet Protocols (IPv4 & IPv6) and subnet and supernet Addressing, Transition from IPv4 to IPv6. Equal-Sized Packets Model: ATM. Address Mapping : Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).		
UNIT II		10 hours
Routing in the Internet : Routing concepts including intra-domain and inter-domain routing, unicast and multicast routing; router architecture, routing tables, packet forwarding, queueing and switching; distance vector routing and RIP; link state routing and OSPF; inter-domain routing using path vector routing and BGP. Internet Control Message Protocol (ICMP) – message types, format, and error reporting; Multicast Routing Protocols: Introduction to IGMP and PIM.		
UNIT III		10 hours
End-to-End Protocols: Transport Layer, Transmission Control Protocol (TCP), User Datagram Protocol (UDP), TCP services, TCP features, Segment, Three way handshaking, Flow control, Error control, Congestion control. Application Layer: HTTP, FTP, Electronic Mail in the Internet, Domain Name System (DNS), P2P File Sharing.		
UNIT IV		10 hours
Advanced Networking : VPNs and tunneling basics; overview of MPLS; introduction to VoIP and SIP with RTP. Wireless Networking: Infrastructure of Wireless Networks, Wireless LAN, IEEE 802.11 standards, Cellular Networks with Mobile IP. Mobile Adhoc Networks: overview of mobile ad hoc networks (MANETs) and basic routing protocols.		
Text Books		
1	C. Siva Ram Murthy and B. S. Manoj, Ad hoc Wireless Networks Architectures and Protocols, 2nd edition, Pearson Education, 2004.	
2	Holger Karl & Andreas Willig, " Protocols and Architectures for Wireless Sensor Networks", John Wiley, Student Edition (Indian), 2016.	
Reference Books		
1	Perkins, Charles, "Ad hoc networking", 1st Edition, Pearson Education India, 2008.	
2	R. Hekmat, "Ad-hoc Networks: Fundamental Properties and Network Topologies", Springer, 1st Edition, 2006	
3	C. K.Toth, "Adhoc Mobile Wireless Networks", Pearson Education; 1st edition, 2015.	
4	W. Dargie and C. Poellabauer, "Fundamentals of Wireless Sensor Networks –Theory and Practice", Wiley, 2010.	

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BIG DATA ANALYTICS

Course Code: BCS-311
Contact Hours: L-3 T-0 P-2
Course Category: DEC

Credits:4
Semester: 6th

Introduction:

Our ability to handle Big Data has increased the strategic value of data. Companies employ Big Data technologies for a wide range of analytics, descriptive, predictive and prescriptive, based on their data assets. Collection, storage and retrieval of data assets and processing them in reasonable response time is crucial today. This course deals with volume, variety and velocity aspects of Big Data. It exposes students to basic techniques for managing and processing such data.

Course Objectives:

At the end of the course students should demonstrate the ability to manage big data and process it.

Pre-Requisites:

Essential: Distributed Systems, Data warehouse

Desirable: NoSQL Databases

Course Outcomes:

CO1: Perform data gathering of large data from a range of data sources.

CO2: Critically analyse existing Big Data datasets and implementations, taking practicality, and usefulness metrics into consideration.

CO3: Understand the role of statistics in the analysis of large of datasets.

CO4: Apply suitable statistical measures and analyses techniques for data of various structure and content and present summary statistics.

Pedagogy:

The course will be delivered in workshop mode with lecture material and problem-solving exercises suitably interspersed during lecture contact hours. Tutorial work shall be pen and paper problem solving as well as coding exercises. Take home work shall be oriented to use of tools



based on lecture content. Students shall install and learn to use these independently. There shall be about 5 hours per week of take-home work.

UNIT 1

Introduction: Types of data, Big Data, Need for Big Data, Big Data: structured and unstructured, Limitations of conventional data management, techniques for handling big data, Big Data Analytics

Data Streams: Real time stream data, issues with stream data, Data stream management system, Windows (time-based, tuple count based), movement of windows: sliding, tumbling, fixed, hopping

Event Streaming: Architecture, events, producers, consumers, use in website activity tracking, stream processing, stream query processing

UNIT 2

Data warehouse for Big Data. Bus, Hub, Scope architecture, Big Data Cluster. Cloud Warehousing, cloud versus on-prem storage

Data Lakes, Lambda and kappa architectures, Meta data management

UNIT 3

Data pipelines: ETL and ELT, streaming data pipelines. Data Governance

Data virtualization: Need for data virtualization, architecture, abstraction, views and services

Hadoop: introduction, HDFS (Hadoop Distributed File System), Hadoop file system interfaces, Flume and scoop

UNIT 4

Map reduce: Architecture, Distributed processing with hadoop, read write programs, mapper, reducer, combiner, partitioner.

HIVE: architecture, features, types of tables

Societal Issues with Big Data: Data rights, policy and regulation, data and ethics, data and communication. Data as a strategic resource

Text Books

1. Gorelik A., The Enterprise Big Data Lake, O'Reilly/Latest Edition

2. Marz N. and Warren J., Big Data: Principles and best practices of scalable realtime data systems, Manning Publications/Latest Edition



3. Erl T. Khattak W., Buhler P., Big Data Fundamentals: Concepts, Drivers & Techniques, The Pearson Service Technology Series from Thomas ERL/Latest Edition

Reference Book

1. DT Editorial Services, Big Data, Black Book, Dreamtech Press/Latest Edition

2. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015

3. Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)

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Data Warehousing and Business Intelligence

Course Code: BAI 314

Contact Hours: L-3 T-0 P-2

Course Category: DEC

Credits: 4

Semester: 6

Introduction: Data warehousing is a method of organizing and compiling data into one database, whereas data mining deals with fetching important data from databases. Data mining attempts to depict meaningful patterns through a dependency on the data that is compiled in the data warehouse.

Course Objective: The objective of the subject is to facilitate the student with the basics of Data Warehouse and Data Mining, to study algorithms and computational paradigms that allow computers to find patterns and regularities in databases, perform prediction and forecasting, and generally improve their performance through interaction with data.

Pre-requisite: Database systems.

Course Outcome: Upon successful completion of the course, students will be able to:

CO1: Understand the distinctive features of database, data warehouse and different schema supported by data warehouses.

CO2: Define different data pre-processing and data quality techniques for data analysis.

CO3: Explain insights, monitor performance and improve decision making.

CO4: Interpret and implement various data mining approaches like association, classification and clustering in real-world domains.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

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CONTENTS

UNIT - I	10 Hours
Introduction to Data Warehousing: Overview, Difference between Database System and Data Warehouse, The Compelling Need for data warehousing, Data warehouse – The building Blocks: Defining Features, data warehouses and data marts, overview of the components, Three tier architecture, Metadata in the data warehouse. Data pre-processing: Data cleaning, Data transformation ETL Process. ETL tools. Defining the business requirements: Dimensional analysis, information packages – a new concept, requirements gathering methods, requirements definition: scope and content.	
UNIT- II	11 Hours
Principles of Dimensional Modelling: Objectives, From Requirements to data design, Multi-Dimensional Data Model, Schemas: the STAR schema, the Snowflake schema, fact constellation schema. OLAP in the Data Warehouse: Demand for Online Analytical Processing, limitations of other analysis methods, OLAP definitions and rules, OLAP characteristics, major features and functions, hyper cubes.	
UNIT-III	11 Hours
OLAP Operations: Drill-down and roll-up, slice-and-dice, pivot or rotation, OLAP models, overview of variations, the MOLAP model, the ROLAP model, ROLAP versus MOLAP, OLAP implementation considerations. Query and Reporting, Executive Information Systems (EIS), Data Warehouse and Business Strategy. Data Mining Basics: What is Data Mining, Data Mining Defined, The knowledge discovery process (KDD Process), Major Data Mining Techniques: Classification and Prediction: Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, KNN Algorithm.	
UNIT- IV	10 Hours
Cluster detection, K- means Algorithm, Outlier Analysis, memory-based reasoning, link analysis, Mining Association Rules in Large Databases: Association Rule Mining, genetic algorithms, neural networks. Data mining tools. Data Mining Applications- The Business Context of Data Mining, Data Mining for Process Improvement, Data Mining as a Research Tool, Data Mining for Marketing, Benefits of data mining	
Text Books	
1	Kamber and Han, —Data Mining Concepts and TechniquesI, Hart Court India P. Ltd. Elsevier Publications Second Edition, 2022/ Latest Edition. .
2	Paul Raj Poonia, —Fundamentals of Data WarehousingI, John Wiley & Sons, 2004/ Latest Edition
Reference Books	
1	W. H. Inmon, “Building the operational data store”, 2nd Ed., John Wiley, 1999/ Latest Edition.
2	Pang- Ning Tan, Michael Steinbach, Anuj Karpatne and Vipin Kumar, Introduction to Data Mining, Pearson, 2021/ Latest Edition.



Multimedia Technologies

Course Code: BCS 312

Contact Hours: L-3 T-0 P-2

Course Category: DEC

Credits: 4

Semester: 6

Introduction: This course is an introduction to the technologies encompassing digital multimedia including images, videos, audios, web pages etc. that allow users to use and communicate with the multimedia over the internet. After learning this course, the student should be able to understand the basic concepts and technologies related to the multimedia, web, hardware and software programs needed to develop and deploy such computer based interactive multimedia applications.

Course Objective:

- To introduce the basic concepts of multimedia data representations and data compression
- To introduce multimedia architecture and databases
- To introduce the various multimedia applications.

Pre-requisite: Programming Concepts, Graphics.

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 To understand the importance of multimedia objects for web application development

CO2 To understand and apply the compression techniques for various multimedia objects

CO3 To understand and study multimedia database systems and content-based information retrieval systems

CO4 To understand the various multimedia applications and latest trends

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.



Cloud Computing Systems and Applications

Course Code: BCS 304

Credits: 3

Contact Hours: L-1 T-0 P-4

Semester: 5

Course Category: SEC

Introduction: Cloud computing is a scalable service provider platform that provides on-demand and pay per use computing service for various types of shared pool of resources such as memory, servers, storage, networking, software, database, applications designing etc., with the help of the internet. This course will introduce various aspects of cloud computing including fundamentals of cloud computing, load balancing techniques, security challenges, case studies and industrial applications of cloud computing. This will help students to use and explore the cloud computing platforms.

Course Objectives:

- To learn the use of various cloud computing services and cloud deployment models.
- Understand the concept of virtualization in cloud computing.
- To apply the concepts of cloud computing for designing, evaluating, simulating and comparing various applications in a cloud computing environment.
- To gain the confidence in resource management and load balancing algorithms in a cloud computing environment.
- To gain the confidence of security attacks and their provisions at various levels of cloud computing. Pre-requisite: Basic understanding of Operating System, Internet, Parallel and Distributed Computing. Course Outcomes:

CO1: To articulate key concepts of cloud computing and computing techniques, strength and limitations of cloud computing with possible application domains.

CO2: To identify the architecture and infrastructure of cloud computing including SaaS, PaaS, IaaS, public cloud, private cloud and hybrid cloud.

CO3: To interpret various data, scalability and cloud services to acquire efficient database for cloud storage.

CO4: To explain the core issues of cloud computing such as security, privacy and interoperability and deal with controlling mechanism for accessing cloud service.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.



UNIT I

08 hours

Cloud Computing Fundamentals: Introduction of cloud computing, History of cloud computing, Trends in computing, Grid computing, Cluster computing, Distributed computing, Utility computing, Fog computing, NIST definition and characteristics of cloud computing, Cloud as green and smart, Cloud service models IaaS, PaaS, SaaS, BPaaS and HaaS etc. SPI framework, SPI vs. traditional IT Model, Cloud deployment models, Benefits and challenges.

UNIT II

07 hours

Virtualization and Cloud Architecture: Virtualization concept, Resource virtualization, Server virtualization, Storage virtualization and Network virtualization, Storage Network Design: Architecture of storage, Analysis and planning, Storage models, Cloud optimized storage, Virtual Box and Microsoft Hyper-V.

UNIT III

08 hours

Cloud Security: Web services, Web 2.0, Web OS, Security challenges and preventive measures, Infrastructure layer, Network layer and Application layer of cloud computing architecture, Security models in cloud, Resource management in cloud computing, Static and dynamic load balancing in cloud computing, Identity access management and Trust in cloud computing. **Cloud Applications:** Cloud for Industry, Cloud Computing for Health care, Cloud computing for Energy Systems, Cloud Computing for Transportation Systems, Cloud Computing for Manufacturing Industry, Cloud computing for Education.

UNIT IV

07 hours

Cloud service providers and case studies: Amazon EC2, Amazon EC service level agreement and recent developments, GoGrid, Salesforce.com, Force.com, Google App Engine, Rackspace, Government of India Cloud, IBM cloud, Eucalyptus cloud. **Migrating into a Cloud:** Introduction, Broad Approaches to migrating into the cloud, the seven- step model of migration in to a cloud.

Text Books:

- 1 B. Sosinsky, "Cloud Computing Bible", 1st Edition, Wiley-India, 2011/ Latest Edition.
- 2 R. Buyya, C. Vecchiola, and S. T. Selvi, "Mastering cloud computing: foundations and applications programming", 1st Edition, Newnes, 2013/ Latest Edition.
- 3 Thomas Erl, Zaigam Mahmood, Ricardo Puttini, "Cloud Computing Concepts, Technology & Architecture", 1st Edition, Pearson India, 2013/ Latest Edition.

Reference Books:

- 1 A. Shawish and M. Salama, "Cloud computing: paradigms and technologies." In Intercooperative collective intelligence: Techniques and applications, Springer, 2014/ Latest Edition.
- 2 M. Miller, "Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online", 1st Edition, Pearson Education India, 2008/



Compiler Design	
Course Code: BIT 321 Contact Hours: L-3 T-0 P-2 Course Category: DCC/DEC	Credits: 4 Semester: 6

Introduction:

This course provides the complete description about inner working of a compiler. This course focuses mainly on the design of compilers and optimization techniques. It also includes the design of Compiler writing tools. This course also aims to convey the language specifications, use of regular expressions and context free grammars behind the design of compiler.

Course Objectives:

- Introduce major concepts of language translation and compiler design.
- Impart the knowledge of practical skills necessary for constructing a compiler.

Prerequisite: Basic programming skills and Theory of Computation

Course Outcomes: The students will be able to:

- CO1:** Explain the compiler architecture and different phases of compilation with compile time error handling.
- CO2:** Compare top down with bottom-up parsers, and develop appropriate parser to produce parse tree representation of the input
- CO3:** Illustrate language tokens using regular expressions, context free grammar and finite automata and design lexical analyzer for a language.
- CO4:** Design a compiler for a small subset of C language.

Pedagogy:

The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.



UNIT I		10 hours
Introduction to compilers – Analysis of the source program, Phases of a compiler, grouping of phases, compiler writing tools– bootstrapping. Lexical Analysis -The role of Lexical Analyzer, Input Buffering, Specification of Tokens using Regular Expressions, Review of Finite Automata, Recognition of Tokens. Syntax Analysis: Review of Context-Free Grammars – Derivation trees and Parse Trees, Ambiguity, Role of parser, Syntax Error Handling, Left Recursion, Left Factoring.		
UNIT II		12 hours
Top-Down Parsing: Recursive Descent parsing, Predictive parsing, LL (1) Grammars. Bottom-Up Parsing: Shift Reduce parsing – Operator precedence parsing (Concepts only). LR parsing – Constructing SLR parsing tables, Constructing Canonical LR parsing tables and Constructing LALR parsing tables. Syntax directed translation: Syntax directed definitions, Bottom- up evaluation of S- attributed definitions, L- attributed definitions, Syntax-Directed Translation Schemes.		
UNIT III		10 hours
Type Checking: Type systems, Specification of a simple type checker. Run-Time Environments: Source Language issues, Storage organization, Storage allocation strategies. Intermediate Code Generation (ICG): Intermediate languages – Graphical representations, Three Address code, Quadruples, Triples. Assignment statements, Boolean expressions.		
UNIT IV		10 hours
Code Generation: Issues in the design of a code generator, Basic Blocks and Flow Graphs, DAG representation of basic blocks, Peephole Optimization, Code generation from DAGs. Code Optimization: Function-Preserving Transformation, Common Subexpression, Copy Propagation, Dead-code Elimination, Loop Optimization, Optimization of Basic Blocks Introduction to Global Data-Flow Analysis.		
Text Books		
1	A. Monica, S. Lam, R. Sethi and D. Ullman, “Compilers – Principles Techniques and Tools”, Pearson Education India; 2 nd edition, 2013/ Latest Edition.	
2	K. C. Louden, “Compiler Construction – Principles and Practice”, Cengage Learning Indian Edition 2006/ Latest Edition.	
Reference Books		
1	A. I. Hollub, “Compiler Design in C”, Pearson Education India; 1 st edition, 2015/ Latest Edition.	
2	A.W. Appel, M. Ginsburg, “Modern Compiler Implementation in C”, Cambridge University Press, 2004/ Latest Edition.	
3	K. Muneeswaran, “Compiler Design”, Oxford University Press, 2012/ Latest Edition.	
4	S. D. Bergmann, “Compiler Design theory, tools and examples”, C/C++ Edition, Rowan University, 2010/ Latest Edition.	

Subject: Advanced Computer Architecture

Course Code: BCS 313
Contact Hours: L-3 T-0 P-2
Course Category: DEC

Credits: 4
Semester: 6

Introduction: This course provides the complete description about the advancements in Computer Architecture. After exploiting the full capacity of execution of uniprocessor system, the speed is enhanced with using multiprocessor and other concepts like pipelining. The algorithms also need to be parallelised for achieving highest speed. This course aims at teaching the complete concepts about the changes in bus system, memory, placements and interconnection of different processors etc.

Course Objective:

- To introduce different concepts for enhancing speed of computation
- To make students learn about different computer architectures.
- To introduce the concepts of parallel algorithms and parallel programming

Pre-requisite: A course on computer organisation, microprocessor, and computer architecture

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 Understand the concept of parallel architecture and Analyse performance metrics

CO2 Understand the concept of pipelining for achieving higher speed.

CO3 Analyse different parallel architectures.

CO4 Gain Confidence in the concepts of parallel algorithms and parallel programming

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I

10 hours

Introduction & Fundamentals: The concept of computer Architecture: Interpretation of concept of computer architecture at different level abstraction, Multi-level hierarchical frame work, description of computer architecture, Introduction to parallel processing: Basic concept, types of level of parallelism, classification of parallel architecture, Basic parallel techniques, relationship between language and parallel architecture. Principles of scalable performance: Performance Metrics and Measures, Speedup Performance Law, Scalability Analysis & approaches, Processor and memory hierarchy: Design Space of Processor, ISA, CISC & RISC, Memory Hierarchy Technology, Virtual Memory Technology

UNIT II

10 hours

Instruction Level Parallel Processor (Parallelism): Pipelined Processors: Basic concept, ILP: Basics, Exploiting ILP, Limits on ILP, design space of pipelines, performance of pipeline, reservation table, And DLX Case Study. VLIW architecture, Superscalar Processor: Super Scalar and superpipeline Design, A case study of ARM 64-bit processor.



UNIT III

10 hours

Data parallel Architecture and MIMD architectures: SIMD Architecture: Design space, fine grain SIMD architecture, coarse grain SIMD architecture, Associative and Neural Architecture, Systolic Architecture, Vector Architectures: Wordlength, vectorization, pipelining, and vector instruction format. Thread and Process Level Parallel Architecture (MIMD Architecture): Multi-threaded Architecture: Design space, computational model, Data flow architecture, hybrid multi shared architecture Distributed memory MIMD, Architecture: Design space, interconnection networks, topology, fine grain system, medium grain system, coarse grain system, Cache Coherence and synchronisation Mechanism Shared memory MIMD Architecture..

UNIT IV

10 hours

Parallel Algorithm and Programming: MPI: Basics of MPI Open MP: Open MP Implementation in 'C', Directives: Conditional Compilation, Internal Control Variables, Parallel Construct, Work Sharing Constructs, Combined Parallel Work-Sharing constructs, Master and Synchronisation Constructs POSIX thread: IEEE POSIX Threads: Creating and Exiting Threads, Simultaneous Execution of threads

Text Books

1. D. Sima, T. Fountain , P. Karsuk , "Advanced Computer Architectures: A Design Space Approach", Pearson Education India; 1st edition, 2002.
2. K. Hwang , N. Jotwani , "Advance Computer Architecture : Parallelism, Scalability, Programmability", McGraw Hill Education; 3rd edition, 2017.

Reference Books

1. Quinn, "Parallel Programming in C with MPI and Open MP", McGraw Hill Edu.; 1st Ed. 2017
2. J. P. Hayes, "Computer Architecture and Organization", McGraw Hill Education; 3rd ed. 2017



BTech 6th semester Syllabus

Wireless Sensor Networks	
Course Code: BCS 314	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: 6
Course Category: DEC	

Introduction: This course introduces the fundamentals of wireless sensor networks (WSNs), including wireless communication principles, network architectures, and protocol design. It provides an understanding of key challenges such as energy efficiency, scalability, and reliability in wireless environments, along with applications of WSNs in real-world scenarios.

Course Objective:

- To understand the fundamentals and architecture of wireless sensor networks.
- To analyze MAC and routing protocols along with design challenges in WSNs.
- To explore energy management, Quality of Service (QoS), and real-world applications of WSNs.

Pre-requisite: Students should have basic knowledge of wireless communication and computer networks.

Course Outcome: Upon successful completion of this course, students will be able to:

CO1: Explain fundamentals of wireless communication and architecture of wireless sensor networks.

CO2: Analyze MAC protocols and standards used in wireless sensor networks.

CO3: Evaluate routing protocols for wireless sensor networks based on design considerations.

CO4: Assess QoS requirements and energy management techniques in wireless sensor networks.

CO5: Describe applications of wireless sensor networks and their practical significance.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

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UNIT I		12 hours
Introduction: Fundamentals of wireless communication technology, electromagnetic spectrum, radio propagation, characteristics of wireless channels, modulation and multiple access techniques.		
Introduction to ad-hoc and wireless sensor networks: key definitions, unique constraints such as energy limitations, advantages, applications of WSNs, design issues, sensor network architecture, data dissemination, and data aggregation.		
UNIT II		10 hours
MAC Protocols: Issues in designing MAC protocols for ad hoc and sensor networks, design goals, classification of MAC protocols, energy efficiency and duty cycling, MAC protocols for sensor networks, location discovery, Quality of Service (QoS), S-MAC, IEEE 802.15.4.		
UNIT III		10 hours
Routing Protocols: Issues in designing routing protocols, classification of routing protocols; table-driven, on-demand, hybrid, flooding, hierarchical, and power-aware routing protocols; cluster-based routing in WSNs; introduction to WSN-specific routing protocols (e.g., LEACH, PEGASIS, TEEN).		
UNIT IV		10 hours
QoS and Energy Management: Issues and challenges in providing Quality of Service (QoS) in wireless sensor networks, MAC and network layer solutions, QoS frameworks; energy management in WSNs, energy-efficient protocols, battery and transmission power management, and energy harvesting techniques.		
Text Books		
1	C. Siva Ram Murthy and B. S. Manoj, Ad hoc Wireless Networks Architectures and Protocols, 2nd edition, Pearson Education, 2004.	
2	Holger Karl & Andreas Willig, " Protocols and Architectures for Wireless Sensor Networks", John Wiley, Student Edition (Indian), 2016.	
Reference Books		
1	Perkins, Charles, "Ad hoc networking", 1st Edition, Pearson Education India, 2008.	
2	R. Hekmat, "Ad-hoc Networks: Fundamental Properties and Network Topologies", Springer, 1st Edition, 2006	
3	C. K.Toth, "Adhoc Mobile Wireless Networks", Pearson Education; 1st edition, 2015.	
4	W. Dargie and C. Poellabauer, "Fundamentals of Wireless Sensor Networks –Theory and Practice", Wiley, 2010.	

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